EVERY POINT OF H OVER AN ABSOLUTELY INTEGRALLY CLOSED ALGEBRA LIFTS TO A POINT OF G

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the translation of the point, the root solving, and the application of the monogenic lemma giving the compatible map.

Editorial note. The label lemma-lift-points-H-to-G names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

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Allowed standard prerequisites (editorial).

- The groups G(R) and H(R) and the monogenic presentation of a point of H. Same chapter, the description of a point of H(R) by an A-algebra map A[t, 1/(pi^ell t + 1)] -> R and of a point of G(R) by A[s, 1/(pi s + 1)] -> R. Supplies the objects between which the map is shown to be surjective.
- The monogenic extension statement for the polynomial Phi. Same chapter, the lemma stating that setting s to a root b of Phi(s) - a gives a compatible A-algebra map. Supplies the unique compatible map producing the lift.

Source reference. Original reference 0GY3: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 14632–14632. Prerequisite pointer invoked at proof line(s) 14692; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let R be an A-algebra which is absolutely integrally closed. Then $G(R) \to H(R)$ is surjective.*

Proof. Let $h \in H(R)$ correspond to the A-algebra map $A[t, \frac{1}{\pi^\ell t+1}] \to R$ sending t to $a \in A$. Since $\Phi(s)$ is monic we can find $b \in A$ with $\Phi(b) = a$. By Lemma 0GY3 sending s to b we obtain a unique A-algebra map $A[s, \frac{1}{\pi s+1}] \to R$ compatible with the map $A[t, \frac{1}{\pi^\ell t+1}] \to R$ above. This in turn corresponds to an element $g \in G(R)$ mapping to $h \in H(R)$. □

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